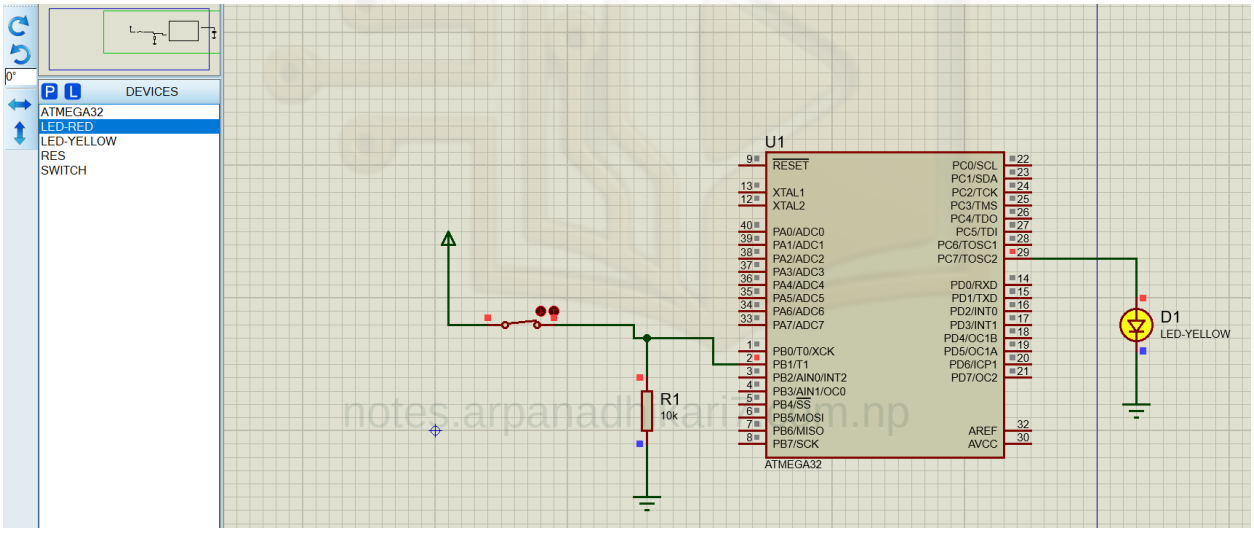
1. A door sensor is connected to bit 1 of port B and LED is connected to bit 7 of port C. Monitor the door sensor, and when it opens, turn the light

Program

```
#include <avr/io.h>    // standard AVR header
int main(void)
{
DDRB = DDRB & 0b11111101;   // pin 1 of Port B is input
DDRC = DDRC | 0b10000000;   // pin 7 of Port C is output
while (1)
{
if (PINB & 0b00000010)  // check pin 1 (2nd pin) of PINB
PORTC = PORTC | 0b10000000;   // set pin 7 (8th pin) of PORTC
else
PORTC = PORTC & 0b01111111;   // clear pin 7 (8th pin) of PORTC
}
return 0;
}
```

Circuit Configuration



2. 7-Segment Display

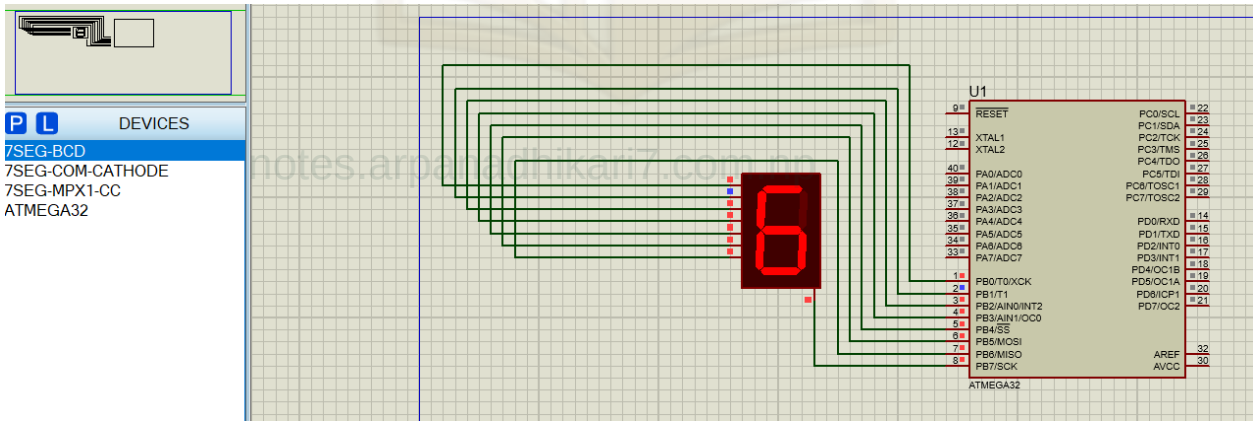
Program

```
#define F_CPU 16000000UL
#include <avr/io.h>
#include <util/delay.h>
unsigned char seg_code[10] = {
0x3F, // 0
0x06, // 1
```

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```
0x5B, // 2
0x4F, // 3
0x66, // 4
0x6D, // 5
0x7D, // 6
0x07, // 7
0x7F, // 8
0x6F  // 9
};
int main(void)
{
DDRB = 0xFF;   // PORTB as output
while(1)
{
for(int i = 0; i < 10; i++)
{
PORTB = seg_code[i];
_delay_ms(100);
}
}
}
```

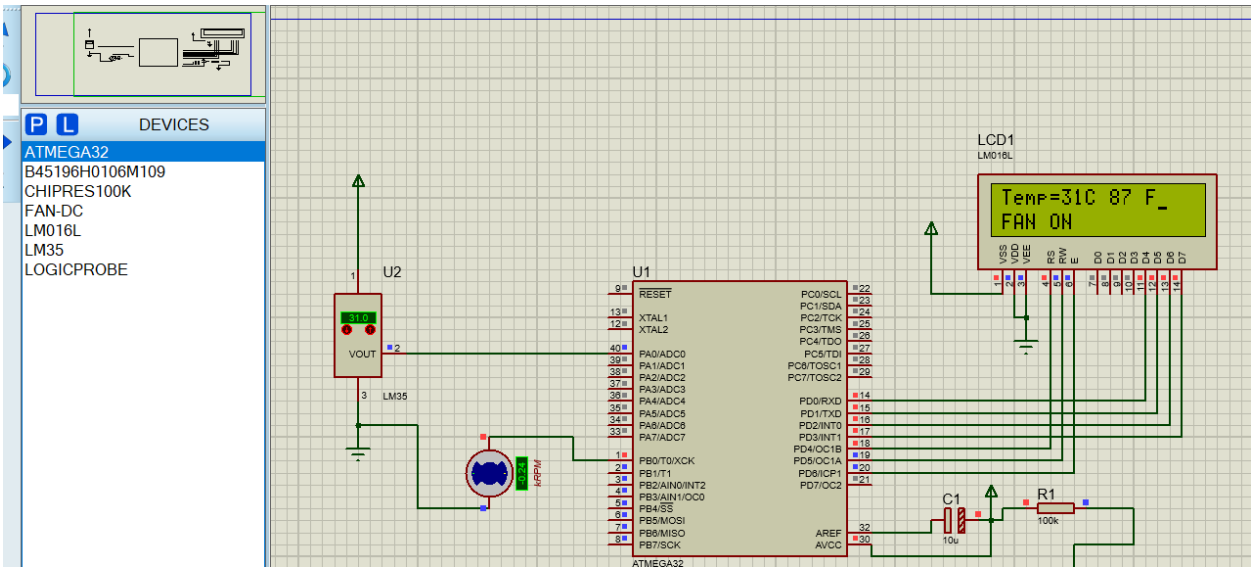
Circuit Configuration



3. Temperature Sensors

Use the folder

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4. LCD with ABCD display

   Program

```
#define F_CPU 16000000UL
#include <avr/io.h>
#include <util/delay.h>

/* LCD Control Pins */
#define LCD_CTRL PORTC
#define LCD_CTRL_DDR DDRC
#define RS PC0
#define RW PC1
#define EN PC2

/* LCD Data Port */
#define LCD_DATA PORTB
#define LCD_DATA_DDR DDRB

void lcd_cmd(unsigned char cmd)
{
        LCD_DATA = cmd;
        LCD_CTRL &= ~(1 << RS);   // RS = 0
        LCD_CTRL &= ~(1 << RW);   // RW = 0
        LCD_CTRL |= (1 << EN);
        _delay_ms(2);
        LCD_CTRL &= ~(1 << EN);
}

void lcd_data(unsigned char data)
```

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```
{
        LCD_DATA = data;
        LCD_CTRL |= (1 << RS);    // RS = 1
        LCD_CTRL &= ~(1 << RW);   // RW = 0
        LCD_CTRL |= (1 << EN);
        _delay_ms(2);
        LCD_CTRL &= ~(1 << EN);
}

void lcd_init()
{
        LCD_DATA_DDR = 0xFF;
        LCD_CTRL_DDR |= (1<<RS)|(1<<RW)|(1<<EN);

        _delay_ms(20);
        lcd_cmd(0x38);   // 8-bit, 2 line
        lcd_cmd(0x0C);   // Display ON
        lcd_cmd(0x06);   // Auto increment
        lcd_cmd(0x01);   // Clear display
        _delay_ms(2);
}

int main(void)
{
        lcd_init();

        lcd_data('A');
        _delay_ms(100);

        lcd_data('B');
        _delay_ms(100);

        lcd_data('C');
        _delay_ms(100);

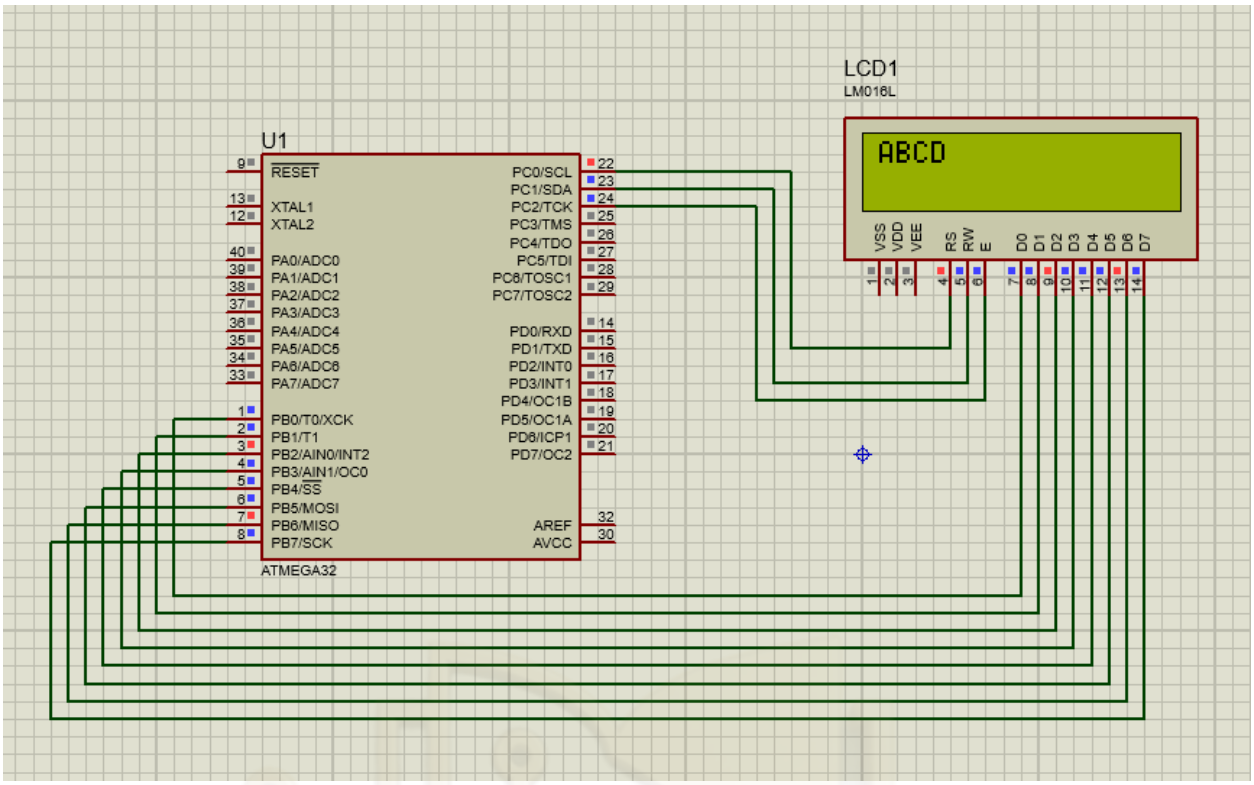
        lcd_data('D');
        _delay_ms(100);

        while (1);
}
```

Circuit Diagram

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5. Your Name on LCD

   Program

```
#define F_CPU 16000000UL
#include <avr/io.h>
#include <util/delay.h>

/* LCD Control Pins */
#define LCD_CTRL PORTC
#define LCD_CTRL_DDR DDRC
#define RS PC0
#define RW PC1
#define EN PC2

/* LCD Data Port */
#define LCD_DATA PORTB
#define LCD_DATA_DDR DDRB

void lcd_cmd(unsigned char cmd)
{
        LCD_DATA = cmd;
        LCD_CTRL &= ~(1 << RS);   // RS = 0
        LCD_CTRL &= ~(1 << RW);   // RW = 0
        LCD_CTRL |= (1 << EN);
```

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```
        _delay_ms(2);
        LCD_CTRL &= ~(1 << EN);
}

void lcd_data(unsigned char data)
{
        LCD_DATA = data;
        LCD_CTRL |= (1 << RS);    // RS = 1
        LCD_CTRL &= ~(1 << RW);   // RW = 0
        LCD_CTRL |= (1 << EN);
        _delay_ms(2);
        LCD_CTRL &= ~(1 << EN);
}

void lcd_string(char *str)
{
        while (*str)
        {
                lcd_data(*str);
                str++;
        }
}

void lcd_init()
{
        LCD_DATA_DDR = 0xFF;
        LCD_CTRL_DDR |= (1<<RS)|(1<<RW)|(1<<EN);

        _delay_ms(20);
        lcd_cmd(0x38);   // 8-bit, 2 line, 5x7
        lcd_cmd(0x0C);   // Display ON, cursor OFF
        lcd_cmd(0x06);   // Increment cursor
        lcd_cmd(0x01);   // Clear display
        _delay_ms(2);
}

int main(void)
{
        lcd_init();

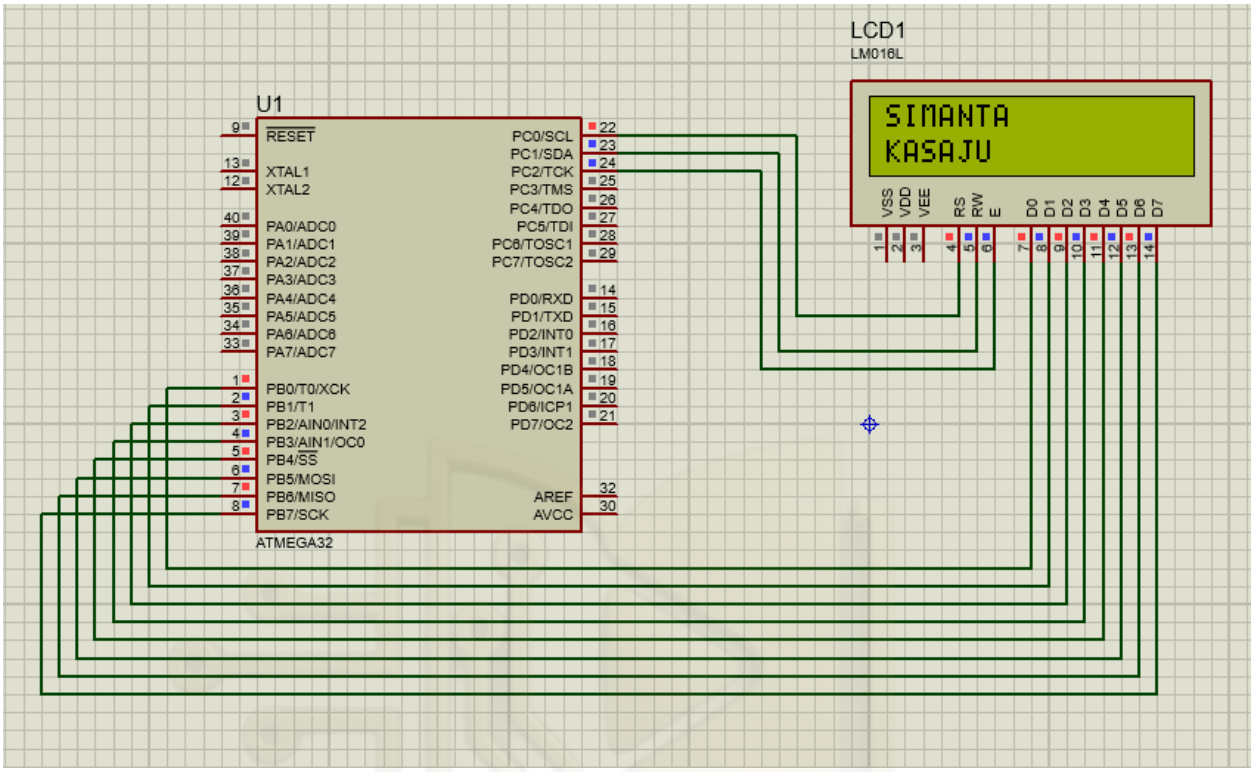
        lcd_cmd(0x80);              // First line
        lcd_string("SIMANTA");

        lcd_cmd(0xC0);              // Second line
```

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```
        lcd_string("KASAJU");

        while (1);
}
```



6. Counter
   Program

```
#define F_CPU 16000000UL
#include <avr/io.h>
#include <util/delay.h>
#include <stdlib.h>   // for itoa

/* LCD Control Pins */
#define LCD_CTRL PORTC
#define LCD_CTRL_DDR DDRC
#define RS PC0
#define RW PC1
#define EN PC2

/* LCD Data Pins */
#define LCD_DATA PORTB
#define LCD_DATA_DDR DDRB
```

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```
/* LCD Functions */
void lcd_cmd(unsigned char cmd)
{
        LCD_DATA = cmd;
        LCD_CTRL &= ~(1 << RS);
        LCD_CTRL &= ~(1 << RW);
        LCD_CTRL |= (1 << EN);
        _delay_ms(2);
        LCD_CTRL &= ~(1 << EN);
}

void lcd_data(unsigned char data)
{
        LCD_DATA = data;
        LCD_CTRL |= (1 << RS);
        LCD_CTRL &= ~(1 << RW);
        LCD_CTRL |= (1 << EN);
        _delay_ms(2);
        LCD_CTRL &= ~(1 << EN);
}

void lcd_string(char *str)
{
        while (*str)
        lcd_data(*str++);
}

void lcd_init()
{
        LCD_DATA_DDR = 0xFF;
        LCD_CTRL_DDR |= (1 << RS) | (1 << RW) | (1 << EN);

        _delay_ms(20);
        lcd_cmd(0x38);   // 8-bit, 2 line
        lcd_cmd(0x0C);   // Display ON
        lcd_cmd(0x06);   // Auto increment
        lcd_cmd(0x01);   // Clear LCD
}

/* Timer1 1-second delay */
void timer1_delay_1s()
{
        TCNT1 = 49911;          // preload for 1 second
        TCCR1A = 0x00;
```

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```
        TCCR1B = (1 << CS12) | (1 << CS10); // Prescaler 1024

        while (!(TIFR & (1 << TOV1))); // Wait for overflow
        TIFR |= (1 << TOV1);          // Clear overflow flag
}

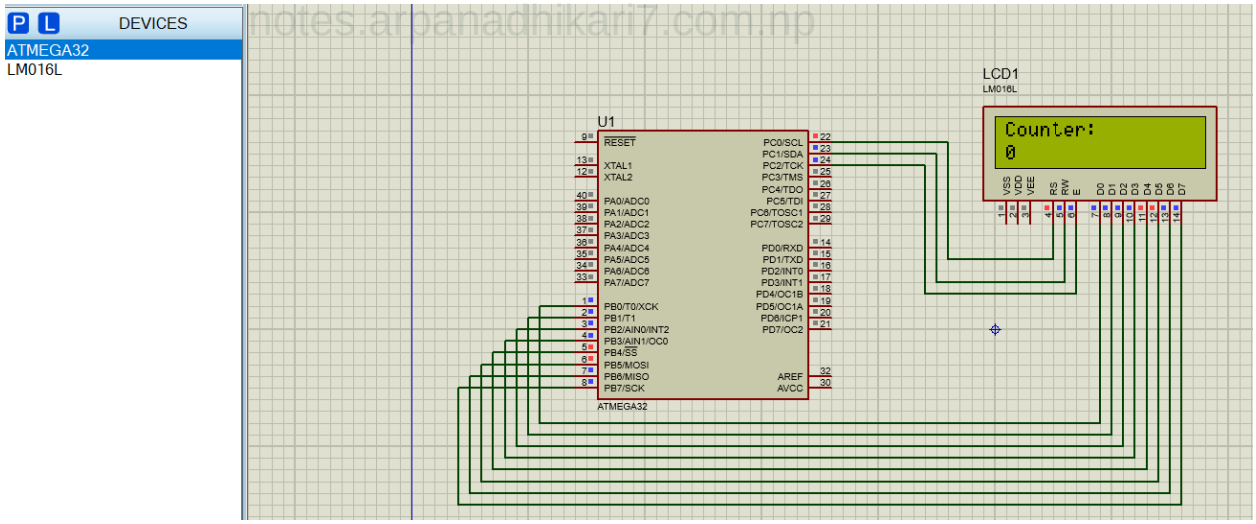
int main(void)
{
        char buffer[6];
        int count = 0;

        lcd_init();
        lcd_cmd(0x80);
        lcd_string("Counter:");

        while (1)
        {
                lcd_cmd(0xC0);          // Second line
                itoa(count, buffer, 10);  // int to string
                lcd_string(buffer);

                count++;
                timer1_delay_1s();
                lcd_cmd(0x01);          // Clear LCD
                lcd_cmd(0x80);
                lcd_string("Counter:");
        }
}
```

Circuit



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